

Use your Class Work from our lesson today to complete the following problems correctly.

Directions: Show your work for the following problems.

- 1) Mr. and Mrs. Jackson invested \$122,500 that they received from the proceeds for the sale of their house into a savings plan that pays 3.5% interest *compounded annually*. The Jacksons intend to leave all of the money in the savings plan for 6 years. How much will their investment be worth at the end of the 6-year period? Find the amount to the *nearest cent*.

$$FV = (122500)(1.035)^6$$

$$FV = 150583.7775$$

$$\text{ANS : } \$150,583.78$$

- 2) Max will purchase a certificate of deposit (CD) for \$12,000. The CD pays 2.8% interest *compounded annually*. Jack has \$12,000 that he will invest in a plan that pays 3% *simple interest*. Which person's investment will be worth more money after 10 years if neither person withdraws any money from his investment?

Max

Jack

$$FV = (12000)(1.028)^{10}$$

$$FV = 15816.57309$$

$$\text{Max : } \$15,816.57$$

$$FV = 1200(0.03)(10) + 1200$$

$$FV = 15600$$

$$\text{Jack : } \$15,600.00$$

Max's investment will be worth more.

- 3) Principal = \$25,750
Rate = $4\frac{1}{2}\%$ APR compounded annually
Time = 8 years

- a. Find the future value of the investment at the end of the investment period.

$$FV = (25750)(1.045)^8$$

$$FV = 36619.09078$$

$$\text{ANS : } \$36,619.09$$

- b. Determine the total amount of interest earned over the investment period.

$$36619.09 - 25750$$

$$\text{ANS : } \$10,869.09$$

- 4) Mr. Harris wrote the following formula for his algebra class to represent the value of a savings account over time (t) in years.

$$f(t) = (1800)(0.025)t + 1800$$

- a. What is the interest rate in Mr. Harris' formula? Give your answer as a percent.

ANS: 2.5%

- b. Is Mr. Harris' formula a simple interest formula or a compound interest formula?

ANS: simple interest

- c. What does 1800 represent in Mr. Harris' formula?

ANS: 1800 is the principal (initial investment).

- d. Use the formula to find $f(5)$.

$$f(5) = (1800)(0.025)(5) + 1800$$

$$f(5) = 2025$$

- e. What does your answer to Part d above mean?

ANS: The answer means that the investment of 1800 at 2.5% simple interest will be worth \$2025 in 5 years.

- f. Describe the graph of Mr. Harris' formula.

ANS: The graph is linear. Its slope is 45 (1800×0.025) and its y-intercept is 1800.

- 5) Ms. Sanchez wrote the following formula for her algebra class to represent the future value of an investment at any rate of interest (r).

$$FV = 14,500(1 + r)^{10}$$

- a. What does 14,500 represent in Ms. Sanchez's formula?

ANS: 14,500 is the principal (initial investment).

- b. What does 10 represent in Ms. Sanchez's formula?

ANS: 10 is the time in years of the investment.

- c. Is Ms. Sanchez's formula a simple interest formula or a compound interest formula?

ANS: compound interest

- d. Use Ms. Sanchez's formula to determine the future value of the investment if interest rate is 8%. Give your answer to the *nearest cent*.

$$FV = (14500)(1.08)^{10}$$

$$FV = 30224.94996$$

ANS : \$30,224.95